

Notes: Campello (JFE, 2003)
Capital structure and product markets interactions: evidence from business cycles

Contents

1	Big picture	3
1.1	Main question	3
1.2	Core results	3
1.3	Why the paper is important	3
2	Theory and hypotheses	4
2.1	General framework: capital structure affects product-market conduct	4
2.2	Long-purse or predation logic	4
2.3	Chevalier–Scharfstein markup mechanism	5
2.4	Strategic commitment theory	5
2.5	Industry equilibrium alternative	5
3	Data and measurement	7
3.1	Industry-level markup data	7
3.2	Markup proxy	7
3.3	Firm-level data	8
3.4	Firm performance measure	8
3.5	Leverage measures	8
3.6	Business-cycle and macro controls	9
4	Models and empirical specifications	9
4.1	Industry-level markup regression	9
4.2	Why first-differencing and 2SLS are used	9
4.3	Firm-level first-stage sales-growth regression	10
4.4	Firm-level second-stage regression	10
4.5	Low-debt versus high-debt industry comparison	11
4.6	Longer-horizon sales effects	11
5	Identification and empirical credibility	12
5.1	What identifies the markup result	12
5.2	What identifies the firm-level result	12
5.3	Why rivals’ leverage matters for interpretation	12
5.4	Potential endogeneity concerns	12
5.5	Robustness structure	13
6	Tables and figures	13

6.1	Table 1: Industry summary statistics	13
6.2	Table 2: Industry debt and markup movements over the business cycle	13
6.3	Figure 1: Cyclical behavior of sales-debt sensitivities	14
6.4	Table 3: Sales-debt sensitivities and the business cycle	14
6.5	Table 4: Robustness to demand cyclicalilty: durables and nondurables	16
7	Section-by-section explanation	16
7.1	Section 1: Introduction	16
7.2	Section 2: Financial structure and product markets	17
7.3	Section 3: Industry debt and markup cyclicalilty	17
7.4	Section 4.1: Firm-level data	17
7.5	Section 4.2: Measuring firm performance	18
7.6	Section 4.3.1: Two-step estimation procedure	18
7.7	Section 4.3.2: Cross-industry comparisons	18
7.8	Section 4.4: Firm-level results	19
7.9	Section 4.5: Economic interpretation	19
7.10	Section 4.6: Robustness	19
7.11	Section 5: Concluding remarks	20
8	Short summary	20
9	Conclusion	21
9.1	Core conclusion	21
9.2	Economic intuition	21
9.3	Takeaway	21

1 Big picture

1.1 Main question

The paper asks whether a firm’s capital structure affects how it competes in product markets, and whether these effects become visible over the business cycle. The central idea is that debt may matter not only because of agency conflicts inside the firm, but also because debt changes how aggressively the firm can compete against rivals when demand is low and external finance is costly.

Key details.

- The empirical object is the interaction between **financial leverage**, **macroeconomic demand shocks**, and **product market outcomes**.
- The paper studies two outcome families:
 - industry-level markup cyclicalities;
 - firm-level relative sales growth, interpreted as a broad measure of market-share performance.
- The main hypothesis is state dependent: leverage should matter more in recessions than in booms.
- The effect should also depend on rivals’ leverage. The same highly levered firm may be vulnerable in a low-debt industry but less vulnerable in a high-debt industry where rivals also face financial constraints.

Interpretation. The paper is about a three-way interaction: own debt, rivals’ debt, and the macro state. This is what makes the design sharper than a simple regression of performance on leverage.

1.2 Core results

- At the industry level, markups are more countercyclical when industry debt is high. In recessions, high-debt industries raise price-cost margins more, or cut them less, than low-debt industries.
- At the firm level, debt hurts relative sales growth in recessions only when rivals are relatively unlevered.
- In low-debt industries, a more levered firm loses market share when aggregate activity weakens.
- In high-debt industries, the same debt-performance relation is not observed; rivals’ financial constraints soften competitive pressure.
- The results are robust to controls for cost shocks, capacity utilization, demand cyclicalities, concentration, monetary policy, default-risk spreads, inflation, alternative lag structures, and instrumental-variable versions of the specifications.

Interpretation. The evidence supports a Chevalier–Scharfstein style mechanism: financially constrained firms invest less in customer acquisition or market-share building during downturns, so they raise markups and lose future sales. The firm-level results also leave room for a state-contingent “long purse” interpretation: financially stronger rivals can take market share from indebted firms when demand is low.

1.3 Why the paper is important

- It connects corporate finance to industrial organization and macroeconomics.
- It gives broad cross-industry evidence rather than focusing on one deregulation or one recession episode.
- It uses business-cycle shocks as a way to reduce the concern that leverage and product-market outcomes are jointly chosen.

- It shows that capital structure can matter for macro dynamics because financing choices alter markup behavior and sales redistribution over the cycle.

Takeaway. Debt is not just a balance-sheet variable. It is also a strategic product-market variable whose effects depend on the state of demand and the financial strength of competitors.

2 Theory and hypotheses

2.1 General framework: capital structure affects product-market conduct

The paper begins from the theoretical literature arguing that financing choices can affect competitive behavior. A firm's capital structure changes the payoff consequences of product-market choices. When debt obligations are large, managers may be forced to focus on short-term cash generation, face tighter refinancing constraints, or shift risk toward creditors. These incentives can influence pricing, output, market-share investment, and survival.

Key details.

- The paper does not estimate a fully structural oligopoly model.
- Instead, it tests broad predictions that several product-market theories of capital structure share.
- The empirical strategy asks when debt should become binding: when demand is low and when rivals are financially stronger.

Interpretation. A cross-sectional relation between leverage and performance can be hard to interpret because leverage may be chosen in anticipation of competition. Campello's contribution is to examine how the relation changes after aggregate shocks under existing financial structures.

2.2 Long-purse or predation logic

The long-purse argument says that financially unconstrained firms can exploit rivals' financial weakness. If a highly levered firm faces a negative shock to demand, its cash flow weakens and debt renegotiation becomes difficult. A rival with a stronger balance sheet can invest in market share, cut prices, or maintain customer acquisition while the levered firm retrenches.

Key details.

- Debt is interpreted as a measure of financial fragility only in specific states.
- The three relevant conditions are:
 1. the firm is highly levered relative to rivals;
 2. the industry is generally low-debt, so rivals are not similarly constrained;
 3. there is a negative demand shock.
- Under those conditions, the indebted firm should lose sales relative to its industry peers.

Interpretation. This is why the paper does not simply compare high-debt firms with low-debt firms. It compares the sales-debt sensitivity across low-debt and high-debt industries and across macroeconomic states.

2.3 Chevalier–Scharfstein markup mechanism

Chevalier and Scharfstein argue that firms can use low prices as an investment in future market share. When liquidity constraints tighten, firms become less willing to sacrifice current profits for future sales. A constrained firm may raise markups during recessions to generate cash today, even if that reduces future customer relationships.

Key details.

- In recessions, externally financed firms are more likely to raise price-cost margins.
- If many rivals are also constrained, all firms raise markups, so no single indebted firm loses much market share.
- If rivals are financially unconstrained, they can keep competing for market share while the indebted firm retrenches.
- Therefore, industry markups should be more countercyclical when industry debt is high, while firm-level market-share losses should be strongest for high-debt firms in low-debt industries.

Interpretation. This is the theory that best matches the combination of results: high industry debt produces more countercyclical markups; high own debt produces sales losses only when rivals are less levered.

2.4 Strategic commitment theory

A different theory, associated with Brander and Lewis, says that debt can make firms commit to aggressive product-market behavior. Because equity is option-like, levered firms may choose more aggressive output policies when limited liability shifts downside risk to debtholders.

Key details.

- Strategic commitment predicts that debt can improve competitive performance.
- If this theory dominates, highly levered firms could gain market share from unlevered rivals.
- The empirical results do not fit a simple unconditional strategic-commitment story because debt hurts sales performance in low-debt industries during recessions.

Interpretation. The paper's results are not that debt is always bad. Instead, debt has state-contingent effects. In bad times and against financially strong rivals, debt appears to constrain competition rather than commit the firm to aggression.

2.5 Industry equilibrium alternative

Industry-equilibrium models emphasize that capital structures and product-market characteristics may be jointly determined. Debt, profitability, scale, technology, and industry structure could all be equilibrium outcomes of deeper industry forces.

Key details.

- This alternative warns against interpreting a static leverage-performance correlation causally.
- Campello's design helps by focusing on macro shocks and by comparing low-debt and high-debt industries in different business-cycle states.

- The specific timing and cross-sectional pattern make it harder to attribute the findings only to equilibrium sorting.

Interpretation. The paper is not a randomized experiment, but the state-contingent and rival-contingent pattern gives the results more credibility than an unconditional capital-structure regression.

3 Data and measurement

3.1 Industry-level markup data

The industry-level analysis studies manufacturing industries at the two-digit SIC level from 1984:IV to 1996:IV. The paper uses industry price data, labor-cost data, output data, capacity utilization, and Compustat-based leverage measures.

Key details.

- Industry prices come from BLS Producer Price Indexes.
- Employment, hours, and wage data come from BLS National Employment, Hours and Earnings.
- Gross output comes from the Bureau of Economic Analysis and is interpolated to quarterly frequency using quarterly GNP.
- Capacity utilization comes from the Federal Reserve's Statistical Release G.17.
- Industry leverage is the asset-weighted average long-term debt-to-asset ratio of Compustat firms in the same two-digit SIC industry.

Interpretation. The markup analysis is designed to capture whether industry-wide financial structure changes pricing behavior over the business cycle.

3.2 Markup proxy

The paper uses a simplified version of Bils's measure of price-cost margins. It adjusts prices for wages, overtime premia, hours, employment, and output, and then removes trend and constants.

$$Markup_t = \ln(P_t/MC_t) = \ln(P_t) - \ln(w_t) - \ln(1 + rV'(H_t)) - \ln(H_t j_t N_t/Y_t) - trend - constant. \quad (1)$$

Key details.

- P_t is price.
- MC_t is marginal cost.
- w_t is hourly average wage.
- r is the legal overtime premium.
- H_t is weekly average hours.
- $V'(H_t)$ captures whether average weekly hours exceed the overtime threshold.
- j_t is the number of weeks in the period, N_t is production workers, and Y_t is gross output.

Interpretation. The markup measure is imperfect because it may not fully control for productivity shifts, but it is attractive because it allows marginal labor cost to move with overtime hours over the cycle.

$$Markup_t = \ln(P_t/MC_t) = \ln(P_t) - \ln(w_t) - \ln(1 + rV'(H_t)) - \ln(H_t j_t N_t/Y_t) - trend - constant, \quad (1)$$

where P is price, w is hourly average wages, r is the legal overtime premium (set to 50% by the Fair Labor Standards Act of 1938), H is weekly average hours, $V(H)$ is

Original paper excerpt: markup measurement equation.

3.3 Firm-level data

The firm-level analysis uses quarterly Compustat data from 1976:I to 1996:IV. The sample is restricted to manufacturing firms in the SIC 200–399 range and to economically meaningful three-digit SIC industries.

Key details.

- Variables: sales, long-term debt, total assets, and plant, property, and equipment (PPE).
- All firm-level variables are inflation-deflated.
- Firms with negative equity or sales growth above 200% in one quarter are excluded.
- Industries with fewer than ten firms in a given industry-quarter are excluded.
- Final sample: 128,133 firm-quarters across 71 three-digit SIC industries.

Interpretation. The firm-level sample is broad enough to study business-cycle interactions across many industries, rather than relying on one industry or one event.

3.4 Firm performance measure

The paper uses a firm’s relative-to-industry sales growth. Campello calls this a proxy for market-share growth, while acknowledging that it is a broad measure.

Key details.

- The dependent variable is the log change in sales, adjusted by industry-specific sales-weighted means.
- The adjustment means the outcome captures how the firm performs relative to competitors.
- The measure is broad: it reflects price, quantity, promotions, investment, service, and other competitive choices.
- It does not identify the exact channel by which the firm competes.

Interpretation. Sales growth is useful because it can be measured consistently across many industries and years. Its disadvantage is that it cannot distinguish whether losses come from pricing, advertising, quality, investment, or customer relationships.

3.5 Leverage measures

At the industry level, leverage is the asset-weighted average long-term debt-to-assets ratio of firms in the two-digit SIC industry. At the firm level, leverage is book long-term debt divided by total assets.

Key details.

- The paper uses book leverage rather than market leverage.
- Book debt is less sensitive to contemporaneous capital-market expectations about future product-market outcomes.
- Long-term leverage levels are interpreted as reflecting cumulative financing decisions rather than immediate responses to sales changes.
- Firm-level right-hand-side variables are industry-adjusted, so the leverage variable is interpreted relative to rivals.

Interpretation. Using lagged and book leverage reduces, but does not eliminate, reverse-causality concerns.

3.6 Business-cycle and macro controls

The empirical strategy uses macroeconomic shocks as changes in the environment in which firms compete.

Key details.

- The main activity proxy in the markup regressions is the negative change in log real GDP, so larger values represent downturns.
- The firm-level second-stage regressions use two activity measures:
 - negative change in log real GDP;
 - change in the unemployment rate multiplied by 100.
- Additional macro controls include changes in the Federal funds rate, the commercial-paper–Treasury-bill spread, and CPI.

Interpretation. These controls separate the real-demand interpretation from monetary-policy, credit-risk, and inflation channels.

4 Models and empirical specifications

4.1 Industry-level markup regression

The markup regression asks whether industry leverage makes markups more countercyclical.

$$Markup_{i,t} = \eta + \alpha[-\Delta \log(GDP)_t] + \beta Leverage_{i,t-1} + \lambda (Leverage_{i,t-1} \times [-\Delta \log(GDP)_t]) + \varepsilon_{i,t}. \quad (2)$$

Key details.

- $Markup_{i,t}$ is the industry markup movement.
- $-\Delta \log(GDP)_t$ is the downturn proxy.
- $Leverage_{i,t-1}$ is lagged industry debt.
- λ is the key coefficient.

Interpretation. If $\lambda > 0$, high-debt industries have markups that rise more when GDP falls. This is the industry-level prediction of the Chevalier–Scharfstein mechanism.

$$Markup_{i,t} = \eta + \alpha(-\Delta \text{Log}(GDP)_t) + \beta Leverage_{i,t-1} + \lambda [Leverage_{i,t-1} \times (-\Delta \text{Log}(GDP)_t)] + \varepsilon_{i,t}. \quad (2)$$

In all estimations of this model, I use Huber–White heteroskedasticity-consistent errors, allowing for within-period error clustering (see [Rogers, 1993](#)).

Original paper excerpt: markup cyclical specification.

4.2 Why first-differencing and 2SLS are used

The paper estimates markup regressions in several ways.

Key details.

- Baseline OLS shows whether markups are countercyclical on average.
- Fixed effects are handled by differencing because industry-specific effects cannot be rejected.
- Differencing may introduce bias if leverage is only weakly exogenous.
- To address this concern, the paper uses 2SLS with lagged leverage levels as instruments.

Interpretation. The robustness of the interaction coefficient across OLS-FE and 2SLS-FE is important. It suggests that the result is not only a fixed-effect or differencing artifact.

4.3 Firm-level first-stage sales-growth regression

The firm-level procedure begins with a cross-sectional regression estimated separately each quarter:

$$\Delta \log(Sales)_{i,t} = \eta + \sum_{k=1}^4 \alpha_k \Delta \log(Sales)_{i,t-k} + \sum_{k=1}^4 \beta_k \Delta \log(PPE)_{i,t-k} + \lambda \log(Assets)_{i,t-1} + \delta_t Leverage_{i,t-1} + \varepsilon_{i,t}. \quad (3)$$

Key details.

- The first-stage coefficient δ_t is the time- t sensitivity of relative sales growth to leverage.
- The dependent variable and regressors are industry-adjusted.
- Four lags of sales growth control for persistence in firm performance.
- PPE growth controls for investment dynamics.
- Lagged assets control for size and borrowing capacity.

Interpretation. The first stage transforms a high-dimensional panel into a time series of leverage-performance sensitivities. A negative δ_t means that more levered firms underperform their rivals in that quarter.

$$\begin{aligned} \Delta \text{Log}(Sales)_{i,t} = & \eta + \sum_{k=1}^4 \alpha_k \Delta \text{Log}(Sales)_{i,t-k} \\ & + \sum_{k=1}^4 \beta_k \Delta \text{Log}(PPE)_{i,t-k} + \lambda \text{Log}(Assets)_{i,t-1} \\ & + \delta_t Leverage_{i,t-1} + \varepsilon_{i,t}. \end{aligned} \quad (3)$$

Original paper excerpt: firm-level first-stage sales-growth regression.

4.4 Firm-level second-stage regression

The second stage regresses the sequence of first-stage leverage coefficients on lagged macro shocks:

$$\delta_t = \eta + \sum_{k=1}^4 \phi_k \Delta Activity_{t-k} + \gamma Trend_t + u_t. \quad (4)$$

Key details.

- $\Delta Activity_{t-k}$ is either a downturn proxy based on GDP or a rise in unemployment.
- The sum $\sum_{k=1}^4 \phi_k$ measures the cumulative effect of a macro shock on the sales-debt sensitivity.
- A negative sum means downturns make the sales-leverage relation more negative.
- Bivariate and multivariate versions add monetary policy, credit spread, and CPI controls.

Interpretation. The second stage is the core business-cycle test. It asks whether recessions make debt more damaging for product-market performance.

$$\delta_t = \eta + \sum_{k=1}^4 \phi_k \Delta Activity_{t-k} + \gamma Trend_t + u_t \quad (4)$$

• use two alternative proxies for negative shocks to activity in all estimations

Original paper excerpt: second-stage time-series regression.

4.5 Low-debt versus high-debt industry comparison

The paper ranks industries by leverage each quarter and compares firms in the bottom and top quintiles.

Key details.

- Low-debt industries: firms in industries in the bottom quintile of industry leverage.
- High-debt industries: firms in industries in the top quintile of industry leverage.
- Eq. (3) is estimated separately for low-debt and high-debt groups.
- Eq. (4) is then estimated separately for each group.
- Differences between the two responses are estimated using a SUR system.

Interpretation. This is the design’s key contrast. If a debt-performance relation appears only in low-debt industries during downturns, then the result is consistent with rivals’ financial strength influencing competitive outcomes.

4.6 Longer-horizon sales effects

The paper also estimates four-quarter-ahead versions of the firm-level procedure, with sales growth computed as

$$\frac{\log(Sales)_{t+3} - \log(Sales)_{t-1}}{4}. \quad (5)$$

Interpretation. This checks whether the short-run debt-performance sensitivity translates into more persistent market-share effects. In the results, four-quarter differences are often larger in magnitude, strengthening the interpretation that debt affects market-share building rather than only one-quarter noise.

5 Identification and empirical credibility

5.1 What identifies the markup result

- The markup analysis uses business-cycle variation and cross-industry differences in leverage.
- The key identifying variation is the interaction of industry debt with aggregate downturns.
- The paper includes industry controls and robustness checks to rule out cost, capacity, concentration, and demand-cyclicality alternatives.

Interpretation. The markup evidence is not purely causal, but the timing around aggregate shocks and the robustness of the interaction term support the view that financial structure affects pricing behavior.

5.2 What identifies the firm-level result

- The firm-level analysis uses macro shocks that affect many industries simultaneously.
- It compares highly levered and less levered firms within the same industry after industry-adjusting variables.
- It then compares the sensitivity across low-debt and high-debt industries.
- The focus is on how the sales-debt relation moves over the business cycle, not just its unconditional level.

Interpretation. The design reduces concerns that some firms are always weaker and always more leveraged. The relevant empirical object is whether recessions make the leverage-performance sensitivity more negative only in certain rival-financing environments.

5.3 Why rivals' leverage matters for interpretation

- If all firms in an industry are highly levered, they may all cut market-share investment in downturns.
- In that case, no single levered firm loses much share because rivals are constrained too.
- If an indebted firm competes against financially stronger rivals, those rivals can continue fighting for market share.
- Thus, own debt matters more in low-debt industries than in high-debt industries.

Interpretation. This cross-industry comparison connects the empirical strategy directly to product-market theory. The firm is not studied in isolation; its debt matters through strategic interaction with rivals.

5.4 Potential endogeneity concerns

- Firms choose leverage, and that choice may reflect expected future product-market conditions.
- Industry debt may be correlated with product-demand cyclicality.
- Markups may move because of productivity or cost shocks, not because of product-market strategy.
- Business-cycle shocks may include both real demand and credit-supply effects.

Interpretation. The paper cannot eliminate all endogeneity. It responds by using lagged and book leverage, industry-adjusted variables, macro shocks, low-debt/high-debt contrasts, fixed effects/differencing, 2SLS, and robustness controls.

5.5 Robustness structure

- Markup robustness: energy-price shocks, capacity utilization, durables/nondurables, concentration, and 2SLS.
- Firm-level robustness: OLS and 2SLS first-stage estimates; GDP and unemployment activity measures; univariate, bivariate, and multivariate second-stage regressions; one-quarter and four-quarter sales horizons.
- Additional checks: alternative lag lengths, alternative monetary-policy measures, and subperiods.
- Demand-cyclical check: repeat firm-level comparisons within durable and nondurable subsamples.

Interpretation. The pattern is robust because the same sign and economic logic appear across multiple empirical designs.

6 Tables and figures

6.1 Table 1: Industry summary statistics

Read:

- The markup sample consists of two-digit manufacturing industries from 1984:IV to 1996:IV.
- Leverage varies substantially across industries: petroleum and electrical machinery have low average leverage around 0.15, while textile mills and stone/clay/glass have high average leverage around 0.35–0.37.
- Capacity utilization varies across sectors and is unavailable for tobacco and instruments.
- Four-firm concentration ratios vary widely, from low concentration in lumber and apparel to high concentration in tobacco and rubber/plastics.
- The table also classifies industries as durable or nondurable following Sharpe’s dichotomy.

Economic message. Industry financial structure is not homogeneous. This cross-industry variation is what allows the paper to test whether leverage affects markup cyclicality.

Table 1
Industry summary statistics. Industries are defined at the two-digit SIC code level and the sample period is 1984:IV–1996:IV. Leverage is computed quarterly as the asset-weighted average debt-to-asset ratio using firm-level data from COMPUSTAT. Capacity utilization is available from the Federal Reserve’s *Statistical Release G-17*. Four-firm concentration ratios are taken from Table 2 in *Rosenberg and Saloner (1986)*. The durable versus nondurable dichotomy is from *Sharpe (1994)*.

SIC	Industry description	Leverage ratio		Capacity utilization (%)		Four-firm (%) concentration ratio	Sharpe’s (1994) dichotomy
		Mean	(Std. dev.)	Mean	(Std. dev.)		
20	Food	0.26	0.03	83.28	0.90	34.50	Nondurable
21	Tobacco	0.27	0.03	N/A	N/A	73.60	Nondurable
22	Textile mills	0.37	0.04	87.21	3.16	34.10	Nondurable
23	Apparel	0.22	0.05	81.43	2.38	19.70	Nondurable
24	Lumber	0.31	0.03	85.59	4.28	17.60	Durable
25	Furniture/fixtures	0.23	0.05	80.88	3.14	21.60	Durable
26	Paper	0.38	0.03	89.46	2.02	31.20	Nondurable
27	Printing/publishing	0.24	0.02	84.21	3.76	18.90	Nondurable
28	Chemicals	0.18	0.01	80.43	2.31	49.90	Nondurable
29	Petroleum	0.15	0.01	88.29	3.49	35.90	Nondurable
30	Rubber and plastics	0.25	0.05	86.10	2.94	60.10	Durable
31	Leather	0.30	0.07	78.79	4.92	24.90	Nondurable
32	Stone, clay, and glass	0.35	0.06	78.31	3.15	37.40	Nondurable
33	Primary metals	0.22	0.02	83.67	6.49	42.90	Durable
34	Fabricated metals	0.27	0.02	78.81	3.35	29.10	Durable
35	Industrial machinery	0.16	0.02	78.93	5.38	36.30	Durable
36	Electrical machinery	0.15	0.02	81.32	3.63	45.00	Durable
37	Transport. Equipment	0.26	0.09	76.34	3.72	50.10	Durable
38	Instruments	0.19	0.03	N/A	N/A	47.80	Nondurable
39	Misc. manufactures	0.20	0.06	75.59	3.81	N/A	Durable

N/A: Not available.

Table 2
Industry debt and markup movements over the business cycle. The dependent variable is based on *Bils (1987)* measure of industry markup movement [see Eq. (1)]. Industries are defined at the two-digit SIC code level and the sample period is 1984:IV–1996:IV. Estimations include a constant term (coefficients omitted). The measure of cyclicality is the negative of the log real GDP. Leverage is computed quarterly for each industry as the asset-weighted average debt-to-asset ratio using firm-level book data. Capacity utilization is the industry quarterly production as a percentage of normal production. Cost shocks are proxied by the change in fuel and energy price index to industry commodity PPI. The durables-nondurables dichotomy is from *Sharpe (1994)*. The “high” four-firm concentration dummy is based on data from *Rosenberg and Saloner (1986)*. Fixed effects (FE) estimations eliminate industry effects by differencing the data. Two-stage least squares (2SLS) are used to control for biases caused by differencing if leverage is only weakly exogenous. The set of instruments includes two lags of leverage (in levels). All estimations use Haber-White heteroskedasticity-consistent *t*-statistics allowing for within-period error clustering.

Estimator	$-\Delta \log(\text{GDP})$	Leverage _{<i>t-1</i>}	$\text{Lev}_{t-1} \times [-\Delta \log(\text{GDP})]$	$\Delta \text{Energy price}$	Capacity util _{<i>t-1</i>}	Durables dummy	Concentration dummy	Adj- <i>R</i> ²	Obs.
1. OLS	10.0764 (5.836)							0.046	924
2. OLS-FE	6.1456 (3.542)	−1.0453 (−2.856)	146.8777 (2.712)					0.035	924
3. OLS-FE*	6.9705 (4.047)	−1.0727 (−2.999)	137.7774 (2.587)	−0.9345 (−3.259)				0.051	832
4. OLS-FE*	7.0551 (4.103)	−1.0447 (−4.638)	128.4153 (3.248)		−0.0047 (−0.709)			0.055	842
5. OLS-FE	6.1307 (3.905)	−1.0479 (−3.266)	146.6471 (2.754)			−0.0016 (−1.120)		0.031	924
6. OLS-FE	6.0988 (3.717)	−1.1362 (−3.441)	151.0229 (2.764)				−0.0015 (−1.070)	0.033	924
7. 2SLS-FE	9.0880 (4.829)	−0.9587 (−1.433)	237.9002 (2.481)					0.036	859

*Excludes SICs 35 and 37.

*Excludes SICs 21 and 38.

6.2 Table 2: Industry debt and markup movements over the business cycle

Read:

- Row 1 shows that markups are countercyclical on average: the coefficient on $-\Delta \log(\text{GDP})$ is 10.0764 with a *t*-statistic of 5.836.

- Row 2 is the main OLS-FE specification: the interaction of lagged leverage with the downturn proxy is 146.8777 with a t -statistic of 2.712.
- Rows 3–6 add controls for energy-price shocks, capacity utilization, durables, and concentration. The interaction remains positive and statistically meaningful.
- Row 7 uses 2SLS-FE and obtains an even larger interaction coefficient of 237.9982 with a t -statistic of 2.481.
- The leverage main effect is negative in most specifications, while the downturn interaction is positive.

Economic message. High-debt industries raise markups more in recessions. This supports the view that financial constraints weaken incentives to cut prices or invest in market share during downturns.

6.3 Figure 1: Cyclical behavior of sales-debt sensitivities

Read:

- The figure plots the cyclical component of first-stage sales-debt sensitivities for low-debt and high-debt industries.
- In low-debt industries, the sales-debt sensitivity becomes more negative during the early-1980s downturn and again around the 1990–91 recession.
- In high-debt industries, the pattern is different and does not display the same pronounced negative downturn movement.
- The descriptive evidence previews the Table 3 result: leverage hurts sales growth in downturns mainly when rivals are not also highly levered.

Economic message. The business cycle changes the competitive meaning of debt. Debt becomes a market-share disadvantage when aggregate conditions worsen and competitors have stronger balance sheets.

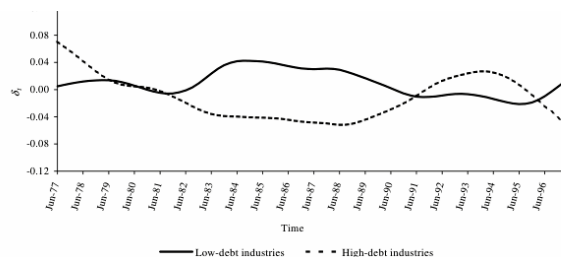


Fig. 1. This figure shows the cyclical behavior of the first-step estimated sales-debt sensitivity of firms in low-debt and those in high-debt industries over the 1977:II–1996:IV period. Quarterly estimates of the sales-debt sensitivities are obtained from the estimation of Eq. (3) in the text via 2SLS. The estimations are performed separately over the two industry subsamples with the coefficients returned for leverage (δ) saved into two separate time series. Firm-quarters are assigned to either low- and high-debt industries conditional on whether their three-digit SIC defined industries were ranked in the bottom or in the top quintile of the industry-wise leverage debt-to-asset ranking, respectively, as of the quarter preceding the estimation of Eq. (3). Cyclical and counter-cyclical components are isolated using the Hodrick and Prescott

Original paper Figure 1: cyclical behavior of sales-debt sensitivities.

6.4 Table 3: Sales-debt sensitivities and the business cycle

Read:

- Panel A uses OLS-estimated first-stage sales-debt sensitivities. Panel B uses 2SLS-estimated sensitivities.
- For low-debt industries, downturns make sales-debt sensitivities negative. In Panel A, GDP-based multivariate estimates give a low-debt sum of -6.300 .
- For high-debt industries, the corresponding sums are typically insignificant and often positive.

- The low-minus-high differences are negative across specifications. In Panel A, the GDP-based multivariate one-quarter difference is -8.135 with a p-value of 0.002; the four-quarter difference is -19.546 with a p-value of 0.001.
- Panel B gives similar or stronger results. GDP-based multivariate estimates show a one-quarter low-minus-high difference of -8.686 with a p-value of 0.002 and a four-quarter difference of -20.904 with a p-value of 0.001.
- Using unemployment instead of GDP gives the same qualitative pattern: low-debt industries show negative sensitivities; high-debt industries do not.

Economic message. When demand weakens, debt depresses relative sales growth only in industries where rivals are relatively less indebted. This is the central firm-level result.

Table 3

Sales-debt sensitivities and the business cycle: two-stage estimator. The dependent variable is the sensitivity of sales growth to leverage [estimated either via ordinary least squares (OLS) or two-stage least squares (2SLS)] for firms operating in industries categorized as low- versus highly-leveraged. In each estimation, the dependent variable is regressed on four lags of a measure of aggregate economic activity, a constant, and a time trend. In the bivariate regressions, the change in the Fed funds rate is added. In the multivariate regressions the changes in Fed funds rate, the change in the CPI, and the change in the spread between the rates on commercial paper and the six-month T-bill are included. The sample period is 1976:I–1996:IV. The sum of the coefficients for the four lags of the economic activity measure is shown along with the p -value for the sum. Exclusion test rows report the p -values for the rejection of the hypothesis that the four lags of the economic activity measure do not forecast the sensitivity of sales growth to debt. Heteroskedasticity- and autocorrelation-consistent errors are computed with a Newey and West (1987) lag window of size four in the one-quarter ahead regressions. The standard errors for the difference of the sum of the four lags of the economic activity measure are computed with a seaming unrelated regression (SUR) system that estimates industry group regressions jointly.

		One-quarter δ			Four-quarters δ
		Low-debt	High-debt	Low – High	Low – High
<i>Panel A: Dependent variable is the OLS-estimated sensitivity of sales growth to leverage</i>					
1. Economic activity measure is the (negative) change of log real GDP					
Univariate	Sum of coefficients	–4.753	1.637	–6.390	–15.016
	Summation test (p -value)	0.009	0.466	0.015	0.004
	Exclusion test (p -value)	0.099	0.588		
Bivariate	Sum of coefficients	–6.345	1.610	–7.955	–19.084
	Summation test (p -value)	0.001	0.471	0.007	0.001
	Exclusion test (p -value)	0.007	0.147		
Multivariate	Sum of coefficients	–6.300	1.835	–8.135	–19.546
	Summation test (p -value)	0.001	0.416	0.002	0.001
	Exclusion test (p -value)	0.013	0.090		
2. Economic activity measure is the change in the rate of unemployment ($\times 100$)					
Univariate	Sum of coefficients	–6.136	0.441	–6.577	–26.273
	Summation test (p -value)	0.128	0.933	0.270	0.029
	Exclusion test (p -value)	0.022	0.164		
Bivariate	Sum of coefficients	–8.713	–0.016	–8.697	–36.967
	Summation test (p -value)	0.020	0.997	0.198	0.009
	Exclusion test (p -value)	0.004	0.130		
Multivariate	Sum of coefficients	–11.420	–0.643	–10.777	–44.666
	Summation test (p -value)	0.003	0.917	0.137	0.006
	Exclusion test (p -value)	0.000	0.818		
<i>Panel B: Dependent variable is the 2SLS-estimated sensitivity of sales growth to leverage</i>					
1. Economic activity measure is the (negative) change of log real GDP					
Univariate	Sum of coefficients	–5.314	1.727	–7.042	–19.460
	Summation test (p -value)	0.002	0.502	0.014	0.001
	Exclusion test (p -value)	0.037	0.333		
Bivariate	Sum of coefficients	–6.250	2.226	–8.476	–21.152
	Summation test (p -value)	0.001	0.380	0.008	0.001
	Exclusion test (p -value)	0.009	0.095		
Multivariate	Sum of coefficients	–6.070	2.616	–8.686	–20.904
	Summation test (p -value)	0.001	0.273	0.002	0.001
	Exclusion test (p -value)	0.021	0.092		
2. Economic activity measure is the change in the rate of unemployment ($\times 100$)					

Table 3 Continued

		One-quarter δ			Four-quarters δ
		Low-debt	High-debt	Low – High	Low – High
Univariate	Sum of coefficients	–6.125	0.268	–5.857	–36.139
	Summation test (p -value)	0.131	0.968	0.383	0.006
	Exclusion test (p -value)	0.091	0.446		
Bivariate	Sum of coefficients	–6.272	0.425	–6.696	–41.226
	Summation test (p -value)	0.088	0.954	0.387	0.007
	Exclusion test (p -value)	0.022	0.506		
Multivariate	Sum of coefficients	10.082	0.340	–10.422	–49.409
	Summation test (p -value)	0.016	0.961	0.198	0.004
	Exclusion test (p -value)	0.040	0.936		

Original paper Table 3: sales-debt sensitivities and business-cycle shocks.

6.5 Table 4: Robustness to demand cyclicity: durables and nondurables

Read:

- Table 4 addresses the concern that low-debt industries may simply be more cyclically sensitive in product demand.
- The paper splits industries into durables and nondurables using Sharpe's classification and repeats the multivariate version of the firm-level test.
- Within durables, GDP-based low-debt coefficients remain negative: -7.606 under OLS and -5.824 under 2SLS.
- Within nondurables, the low-minus-high GDP differences are also negative, especially under 2SLS where the difference is -8.219 with p-value 0.056.
- Using unemployment also gives negative low-minus-high differences, with the nondurables 2SLS difference especially large at -20.250 .
- The signs and magnitudes resemble the main Table 3 evidence.

Economic message. The main result is not just a disguised durable-versus-nondurable demand-cyclicity effect. Even after conditioning on demand cyclicity, leveraged firms in low-debt industries are more vulnerable in downturns.

Table 4
Sales-debt sensitivities and the business cycle subsamples. The dependent variable is the one-quarter sensitivity of sales growth to leverage ratio (estimated either via ordinary least squares (denote it by δ_{OLS}) or two-stage least square (δ_{2SLS}), for firms operating in industries categorized as low- versus highly-leveraged. In each estimation, the dependent variable is regressed on four lags of a measure of aggregate economic activity, four lags of the change in the Fed funds rate, four lags of the change in the CPI, four lags of the change in the spread between the rates on commercial paper and the six-month T-bill, a constant, and a time trend. The sample period is 1970:1–1996:IV. The sum of the coefficients for the four lags of the economic activity measure is shown along with the p-value for the sum. Exclusion test rows report the p-values for the rejection of the hypothesis that the four lags of the economic activity measure do not forecast the sensitivity of sales growth to debt. Heteroskedasticity- and autocorrelation-consistent errors are computed with a Newey and West (1987) lag window of size four. The standard errors for the difference of the sum of the four lags of the economic activity measure are computed with a seemingly unrelated regression (SUR) system that estimates industry group regressions jointly. Durables and nondurables industry definitions are based on Sharpe (1994).

		δ_{OLS}			δ_{2SLS}		
		Low-debt	High-debt	Low – High	Low-debt	High-debt	Low – High
Panel A: Economic activity measure is the (negative) change of log real GDP							
1. Durables	Sum of coefficients	-7.606	-0.713	-6.894	-5.824	1.022	-6.846
	Summation test (p-value)	0.002	0.804	0.054	0.012	0.728	0.088
	Exclusion test (p-value)	0.016	0.155	0.136	0.156	0.145	0.088
2. Nondurables	Sum of coefficients	-2.085	3.846	-5.931	-2.743	5.474	-8.219
	Summation test (p-value)	0.227	0.214	0.107	0.093	0.182	0.056
	Exclusion test (p-value)	0.108	0.098	0.099	0.099	0.111	0.111
Panel B: Economic activity measure is the change in the rate of unemployment ($\times 100$)							
1. Durables	Sum of coefficients	-15.878	-6.400	-9.478	-8.641	-3.095	-5.546
	Summation test (p-value)	0.000	0.438	0.342	0.030	0.740	0.628
	Exclusion test (p-value)	0.000	0.421	0.076	0.076	0.311	0.076
2. Nondurables	Sum of coefficients	-3.936	5.571	-9.508	-6.857	13.393	-20.250
	Summation test (p-value)	0.524	0.529	0.346	0.295	0.227	0.093
	Exclusion test (p-value)	0.000	0.480	0.014	0.014	0.345	0.014

Original paper Table 4: sales-debt sensitivities in durable and nondurable subsamples.

7 Section-by-section explanation

7.1 Section 1: Introduction

The introduction frames the paper as a response to a measurement problem in empirical corporate finance. Many theories say debt changes product-market competition, but static correlations between leverage and performance are difficult to interpret because leverage and performance may be jointly determined by industry trends, investment opportunities, capacity, and expected demand. Campello therefore studies how leverage-performance relations change after aggregate shocks, and how those changes depend on rivals' leverage.

Key details.

- Main empirical claim: debt-financed firms underperform in recessions when rivals are unlevered.

- Main industry claim: high-debt industries have more countercyclical markups.
- Main interpretation: financing choices affect cyclical competitive dynamics.
- Main contribution: the paper provides both firm-level and industry-level evidence across many industries and many years.

Takeaway. The introduction sets up the paper’s defining idea: capital structure has strategic external effects because debt changes how firms respond to macro shocks and how rivals can exploit those responses.

7.2 Section 2: Financial structure and product markets

This section reviews product-market theories of capital structure and translates them into testable implications. The long-purse view predicts that indebted firms are vulnerable when financially stronger rivals can continue competing. The Chevalier–Scharfstein view predicts that constrained firms raise markups in recessions because they cannot afford to invest in future sales. The strategic-commitment view gives a contrasting prediction: debt can commit a firm to aggressive behavior. Industry-equilibrium models warn that leverage may reflect deeper industry characteristics rather than causal effects.

Key details.

- The paper does not claim to fully distinguish every theory.
- It focuses on predictions that involve recessions and rivals’ balance sheets.
- The most diagnostic prediction is that debt-induced market-share loss should be strongest when rivals are relatively unlevered.

Takeaway. Section 2 justifies the empirical partitions. Low-debt and high-debt industries are not arbitrary groups; they represent different rival-financing environments.

7.3 Section 3: Industry debt and markup cyclicalities

Section 3 constructs industry-level markup measures and tests whether high-debt industries have more countercyclical markups. The markup proxy is based on Bils’s method, and the key explanatory variable is the interaction between lagged industry leverage and a downturn proxy. The main result is that the interaction is positive and robust.

Key details.

- A 1% GDP decline raises markups on average.
- The high-debt interaction means that markups rise more in industries with greater reliance on debt.
- Robustness checks address input-cost shocks, utilization, durable demand, concentration, and weak exogeneity of leverage.

Takeaway. This section provides industry-level evidence for the pricing side of the mechanism: financially constrained industries behave less aggressively in recessions, raising markups when demand weakens.

7.4 Section 4.1: Firm-level data

This subsection describes the firm-level data construction. The paper uses quarterly Compustat manufacturing firms from 1976:I to 1996:IV, restricted to economically meaningful three-digit SIC industries and filtered for data quality.

Key details.

- The sample contains 128,133 firm-quarters across 71 three-digit SIC industries.
- Variables are inflation-deflated.
- The analysis excludes extreme sales-growth observations and sparse industry-quarters.

Takeaway. The firm-level data enable the paper to test how leverage affects relative performance across many product markets and business-cycle episodes.

7.5 Section 4.2: Measuring firm performance

The paper chooses relative sales growth as the product-market performance measure. This is broader than markup behavior because firms can compete through advertising, service, output, investment, product quality, and promotions.

Key details.

- Relative sales growth is a proxy for market-share performance.
- It is industry-adjusted to remove common industry shocks.
- It is not a perfect market-share measure because Compustat does not cover all firms, especially very small firms.

Takeaway. The performance measure sacrifices mechanism-specific precision for broad coverage and comparability.

7.6 Section 4.3.1: Two-step estimation procedure

The two-step procedure first estimates time-specific sensitivities of relative sales growth to leverage, then studies how those sensitivities respond to macro shocks.

Key details.

- Step 1 estimates δ_t , the sales-growth sensitivity to leverage in each quarter.
- Step 2 regresses δ_t on lagged macro activity shocks.
- The method is less efficient than a one-step panel regression, but it avoids forcing coefficient stability across time and industry groups.
- Newey–West standard errors address autocorrelation and heteroskedasticity in the second stage.

Takeaway. The two-step method is designed to study changing sensitivities rather than fixed average effects.

7.7 Section 4.3.2: Cross-industry comparisons

The paper’s most important identification move is to compare low-debt and high-debt industries. If debt simply reflects weak firms, then results should not line up so tightly with rivals’ debt. If product-market theory is correct, debt should hurt most when rivals are financially strong.

Key details.

- Industries are ranked each quarter by leverage.
- Firms in the bottom quintile are assigned to low-debt industries.
- Firms in the top quintile are assigned to high-debt industries.
- Separate first-stage and second-stage regressions are estimated for each group.

Takeaway. This subsection turns rivals' financing into the central conditioning variable.

7.8 Section 4.4: Firm-level results

Table 3 shows that downturns make the sales-debt sensitivity negative in low-debt industries but not in high-debt industries. This result appears under both GDP and unemployment proxies, under OLS and 2SLS first-stage estimates, and at one-quarter and four-quarter horizons.

Key details.

- In low-debt industries, downturns predict more negative leverage-performance sensitivity.
- In high-debt industries, the sensitivity is not significantly negative.
- Low-minus-high differences are negative in all pairs and statistically significant in most cases.

Takeaway. The firm-level results are the clearest evidence that product-market consequences of debt depend on rivals' financial positions.

7.9 Section 4.5: Economic interpretation

The paper translates coefficients into economic magnitudes. A firm whose debt-to-asset ratio is 10 percentage points above the industry average is compared with a rival 10 percentage points below the average. In the multivariate GDP specification, following a 1% GDP decline, the more indebted firm in a low-debt industry grows sales roughly 1.3% less than the less indebted rival. Over four quarters, the low-debt industry underperformance relative to high-debt industries is also economically meaningful.

Key details.

- Coefficients are not just statistically significant; the implied sales-growth differences are large relative to industry sales growth responses to GDP.
- The four-quarter horizon indicates persistent effects.
- The magnitudes fit a market-share-building interpretation.

Takeaway. The effects are large enough to matter for product-market strategy and for macroeconomic fluctuations.

7.10 Section 4.6: Robustness

The robustness section addresses the concern that low-debt industries may be more cyclical in product demand. Campello splits industries into durables and nondurables based on Sharpe's classification and repeats the analysis.

Key details.

- Results remain negative for low-debt industries within both durables and nondurables.
- High-debt coefficients remain insignificant.
- Difference coefficients retain the expected sign and similar magnitudes.
- Alternative lag structures, monetary-policy measures, and subperiods also yield qualitatively similar results.

Takeaway. Robustness checks strengthen the argument that the main effect is about financial structure and strategic interaction, not simply product-demand cyclicity.

7.11 Section 5: Concluding remarks

The conclusion emphasizes that capital structure has systematic product-market implications. Debt affects firm performance in downturns and influences industry markup behavior.

Key details.

- Debt has negative firm-level effects in low-demand states when rivals are less constrained.
- Markups are more countercyclical when industry debt is high.
- Financing decisions have implications for product-market dynamics and macroeconomic fluctuations.

Takeaway. The paper's broad lesson is that capital structure can transmit and amplify business-cycle shocks through product-market competition.

8 Short summary

- Campello studies whether debt affects product-market behavior over the business cycle.
- The paper combines industry-level markup tests and firm-level sales-growth tests.
- The theory comes mainly from long-purse/predation models and Chevalier-Scharfstein models of financially constrained market-share investment.
- Industry markups are more countercyclical when industry leverage is higher.
- Firm-level relative sales growth is hurt by debt in recessions only in low-debt industries.
- The mechanism is rival-dependent: a levered firm is more vulnerable when rivals have stronger balance sheets.
- The main firm-level method estimates sales-debt sensitivities each quarter and then asks whether macro downturns make those sensitivities more negative.
- Low-debt industries show negative responses to GDP declines and unemployment increases; high-debt industries do not.
- Four-quarter results are stronger, suggesting persistent market-share consequences.
- Robustness checks address cost shocks, capacity utilization, concentration, product-demand cyclicity, weak exogeneity, monetary policy, default-risk spreads, and alternative lag structures.
- The main limitation is that the paper is still observational and uses broad proxies for demand shocks and product-market performance.
- The central contribution is to show that capital structure affects competition in a systematic, state-contingent way.

9 Conclusion

9.1 Core conclusion

Campello shows that capital structure affects product-market outcomes, especially during downturns. High industry debt is associated with more countercyclical markups, and high firm debt is associated with worse relative sales growth when rivals are less indebted.

9.2 Economic intuition

Debt is costly in bad states because it limits a firm's ability to invest in customers, lower prices, or maintain market share. If competitors are also highly levered, they face similar constraints, so the competitive penalty is muted. If competitors are financially stronger, they can exploit the indebted firm's weakness.

9.3 Takeaway

The paper's enduring message is that debt changes competition. Financing choices influence not only payouts and investment inside the firm, but also how the firm behaves toward rivals and customers across the business cycle.